| Table 2.2 Classification of Algae | | | |
|---|---|---|---|
| **Class** | **Pigments** | **Flagella** | **Reserve food** |
| Chlorophyceae | Chlorophyll a and b Carotenoids, Xathophyll | 1,2,4 or more equal anterior whiplash flagella | Starch |
| Xanthophyceae | Chlorophyll a and b Carotenoids Xathophyll | 2, unequal anterior 1 tinsel and 1 whiplash | Fats and leucosin |
| Chrysophyceae | Chlorophyll a and b Carotenoids, | 1 or 2 unequal or equal anterior both whiplash or 1 whiplash and 1 tinsel | Oils and leucosin |
| Bacillariophyceae | Chlorophyll a and c Carotenoids, | 1 anterior (only in male gametes) tinsel | Leucosin and Fats |
| Cryptophyceae | Chlorophyll a and c carotenoids and xanthophyll | unequal anterior both tinsel flagella | Starch |
| Dinophyceae | Chlorophyll a and c carotenoids and xanthophyll | Two unequal (whiplash)lateral flagella in different plane | Starch and oil |
| Chloromonadineae | Chlorophyll a and b Carotenoids, Xathophyll | 2 equal flagella | oil |
| Euglenophyceae | Chlorophyll a and b | One or two anterior tinsel flagella | Fats and paramylon |
| Phaeophyceae | Chlorophyll a and c, Xanthophyll | Two unequal whiplash and tinsel lateral flagella | Laminarin starch and fats |
| Rhodophyceae | Chlorophyll a, r-Phycoerthythrin | absent | Floridean starch |
| Cyanophyceae | Chlorophyll a, carotenoids, c-Phycocyanin,Allophycocyanin | absent | Cyanophycean starch |

and *Macrocystis*). The thallus is differentiated into leaf like photosynthetic part called fronds, a stalk like structure called stipe and a holdfast which attach thallus to the substratum.

The Pigments include Chlorophyll a, c, Carotenoids and Xanthophylls. A golden brown pigment called fucoxanthin is present and it gives shades of colour from olive green to brown to the algal members of this group. Mannitol and Laminarin are the reserve food materials. Motile reproductive structures are present. Two laterally inserted unequal flagella are present. Among these one is whiplash and another is tinsel. Although sexual reproduction ranges from isogamy to oogamy, Most of the forms show oogamous type. Alternation of generation is present (isomorphic, heteromorphic or diplontic). Examples for this group include *Sargassum, Laminaria, Fucus* and *Dictyota.*

### Rhodophyceae

Members of this group include '**Red algae**' and are mostly marine. The thallus is multicellular, macroscopic and diverse in form. *Porphyridium* is the unicellular form. Filamentous (*Goniotrichum*) ribbon like (*Porphyra*) are also present. *Corallina* and *Lithothamnion* are heavily impregnated with lime and form coral reefs. Apart from chlorophyll a, r-phycoerythrin and r-phycocyanin are the photosynthetic pigments. Asexual reproduction takes place by means of monospores, neutral spores and tetraspores.

The storage product is floridean starch. Sexual reproduction is oogamous. Male sex organ is spermatangium which produces spermatium. Female sex organ is called carpogonium. The spermatium is carried by the water currents and fuse with egg nucleus to form zygote. The zygote develops into carpospores. Meiosis occurs during carpospore formation. Alternation of generation is present. Examples for this group



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